Michelle Liu

(775) 772-8748 | michellemq6@gmail.com | michelleliu-4 | michelleliu4 | michelleliu4.github.io

EDUCATION

Brown University Providence, RI
Bachelor of Science, Applied Math – Computer Science, GPA: 4.00/4.00 *Sep 2021 – May 2025*

- **Awards:** Kleiner Perkins Fellow, Robinhood Hackathon Bronze, AIME Qualifier, Rohn Truell Prize in Applied Mathematics

EXPERIENCE

Databricks Oct 2025 – Present
Software Engineer *Mountain View, CA*

- Architected and led a frontend refactor for Databricks Assistant's tool-calling system, migrating *60+ agentic tools* to a new **React**/**TypeScript** framework and redesigning interaction surfaces for data workflows
- Optimized agent performance and reliability by implementing a TTL-based state management system, enhancing notebook context-passing logic, and maintaining a suite of **Playwright** tests and **Grafana** probes to monitor UI health

Figma Jun 2025 – Aug 2025
Software Engineer Intern *San Francisco, CA*

- Developed custom stroke endcap sizing feature within Figma's Core Experience team, with **C++**, **React**, and **TypeScript**
- Engineered geometry and SceneGraph logic to enable precise, user-controlled cap sizes through vector math
- Collaborated with designers to iterate the stroke settings panel design, shaping the endcap controls and overall **UI/UX**

Lightspeed Jan 2024 – May 2025
Data Science Intern *Providence, RI*

- Architected an LDA-based restaurant cohorting model to segment *5,000+ businesses*, optimizing menu pricing elasticity
- Analyzed transactions using **BigQuery** to uncover sales patterns during holidays, informing restaurant strategy
- Developed sales and tip trend analyses using **Google Cloud Platform** Dataform and **SQL**, with Looker for strategic planning
- Led initiative to optimize table pacing, leveraging statistical models to support just-in-time alerts (**scikit-learn, pandas**)

Susquehanna International Group Jun 2024 – Aug 2024
Quantitative Trading Intern *New York, NY*

- Quantified the correlation between market attention and stock volatility using time-series analysis and hypothesis testing with **Python** and **riptable**
- Executed real-time trading strategies and risk management in simulated **electronic mock trading** sessions

Brown University Aug 2022 – Dec 2023
Undergraduate Teaching Assistant *Providence, RI*

- Assisted over *300 students* and designed over *10 assignments* for **NLP**, statistics, and **machine learning** courses

JPMorgan Chase & Co. Jun 2023 – Aug 2023
Software Engineer Intern *Jersey City, NJ*

- Engineered **Splunk**/**AWS** pipelines to preprocess call center logs for anomaly detection and regional trend analysis
- Built outlier and time-series models (*90%+ accuracy*) to identify irregular call patterns for *10,000+ specialists*

PROJECTS

Jreamboard | *React, Node.js, JavaScript, Express*

- Developed a podcast and audio-based social media web application for the Jream Foundation
- Prototyped frontend using **Figma** and implemented UI using **React**, designing a login and audio posts page
- Built backend architecture with **Express** and **PostgreSQL** to store audio and account information

IntelliGin | *Python*

- Developed a Double DQN (Deep Q-Network) **reinforcement learning** model for Gin Rummy, employing experience replay, self-play and adversarial strategies, achieving performance on par with rule-based adversaries and random agents
- Implemented custom environment in RLCard for Gin Rummy, including engineered feature extraction

Caching I/O | *C*

- Designed caching system for file reading and writing, achieving *31% faster* performance than C's standard library

North Burial Ground | *R, HTML/CSS, Adobe Illustrator*

- Partnered with Providence government to publish historical burial trends at historic cemetery using **ggplot2** and Illustrator

SKILLS

Languages: Python (proficient); C, C++, SQL, JavaScript, TypeScript, HTML/CSS (intermediate); R (novice)
Frameworks & Tools: React, Figma, Grafana, Playwright, scikit-learn, pandas, TensorFlow, PyTorch, GCP, AWS